ClassQuest — User Guide

Cloud Computing Prototype · Student, Teacher and Administrator Manual

ClassQuest Project (INFS803)

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Chapter 1

About this guide

This is the end-user manual for **ClassQuest**, a lightweight online course platform. It explains how to sign in, how each kind of user works day to day, and how to get the system running on your own machine for a demonstration.

ClassQuest has three kinds of users:

- **Students** browse published courses and open their resources (documents, digital books and videos), and track which ones they have opened.
- **Teachers** create courses, upload resources into them, and publish courses so students can see them. Teachers also have access to an **Operations** screen for demonstrations.
- **Administrators** monitor the health of the system, run the same operations controls, and browse every resource in the library.

  This is a research prototype. All content is synthetic sample data, and the system is meant to be run locally for demonstration. It is not a production service and does not store real student records, grades or personal data.

Who should read which part:

You are a…	Start at
Student	Getting started → Student guide
Teacher	Getting started → Teacher guide
Administrator	Getting started → Administrator guide
Person setting up a demo	Running ClassQuest locally

Chapter 2

Key concepts

A few terms are used throughout this guide.

- **Course** — a container for related learning resources. A course has a title, a description, a category, an optional cover image, and a status.
- **Resource** — a single learning item inside a course. There are three types: **Document**, **Book** (a digital book) and **Video**. Each resource can carry an optional week or section label.
- **Course status** — a course is one of:
  - **Draft** — only its teacher (and administrators) can see it. Students cannot.
  - **Published** — visible to students.
  - **Archived** — hidden from students again; its resources and history are kept, but no new resources can be added.
- **Resource processing** — when a resource is uploaded, it goes through a short automatic pipeline (**Submitted** → **Queued** → **Processing** → **Completed**) before it is ready. This usually takes a few seconds. Students only ever see resources that have reached **Completed**.
- **My Progress** — for students, this means *resources opened*. It is a record of what you have opened, overall and per course. It is **not** a grade, a score, or a measure of mastery or lesson completion.

Chapter 3

Getting started

3.1 Signing in

1. Open ClassQuest in your web browser at **http://localhost:8080**.
2. Enter your email and password on the **Login** screen.
3. Select **Sign in**. You land on the home screen for your role:
    - Students land on **Home**.
    - Teachers land on their **Dashboard**.
    - Administrators land on **Operations**.

3.1.1 Demonstration accounts

When the prototype runs in demo mode (the default), three accounts are created for you automatically. These are public, synthetic credentials for demonstration only.

Role	Email	Password
Student	`student@classquest.example`	`DemoStudent123!`
Teacher	`teacher@classquest.example`	`DemoTeacher123!`
Administrator	`admin@classquest.example`	`DemoAdmin123!`

3.2 Finding your way around

Every screen shares the same layout:

- A **sidebar** on the left with the pages available to your role.
- A **top bar** showing the current page and your account, with the option to sign out.

The pages you see depend on your role:

Role	Pages in the sidebar
Student	Home · Courses · My Progress
Teacher	Dashboard · My Courses · Publish Resource · Operations

Role	Pages in the sidebar
Administrator	Operations · Library

If you try to open a page that is not meant for your role, ClassQuest simply returns you to your own home page.

3.3 Signing out

Use the account control in the top bar to sign out. Your session is secured with a sign-in token that expires automatically after about an hour, so you may be asked to sign in again during a long session.

Chapter 4

Student guide

As a student you can browse published courses, open their resources, and see your own progress. You cannot see draft or archived courses.

4.1 Home

The **Home** screen is your starting point. When you have opened at least one resource, it shows a **Continue where you left off** shortcut so you can jump straight back to recent material.

4.2 Browsing courses

1. Select **Courses** in the sidebar.
2. You see every **published** course as a card, with its cover image (or a coloured placeholder), title, category, and your own progress for that course (for example *3 / 4 resources opened · 75%*).
3. Select a course to open it.

Draft and archived courses are never listed for students, so the list only ever shows material a teacher has deliberately published.

4.3 Opening a resource

Inside a course:

1. Resources are grouped by their **week or section label** and shown in the order the teacher arranged them.
2. Only **completed** resources appear. If a teacher has just uploaded something, it shows up once processing finishes.
3. Select **Open** on a resource. ClassQuest opens the file in a new browser tab using a secure, time-limited link (valid for about five minutes). If you leave the tab open for a long time and the link expires, simply select **Open** again for a fresh link.

4.4 Tracking your progress

Select **My Progress** in the sidebar to see:

- Your **overall** totals — how many resources you have opened out of how many are available to you.
- A **coverage** percentage.
- A breakdown **by resource type** (documents, books, videos).
- A breakdown **by course**, each with opened / available and a percentage.
- A **Recently opened** list.

Remember: progress counts a resource the moment you open it. It reflects what you have *looked at*, not how well you understood it. Only completed resources in published courses are counted, so your "opened" total can never exceed your "available" total.

Chapter 5

Teacher guide

As a teacher you create courses, add resources, arrange and publish them, and manage each course through its lifecycle. You can only manage courses **you** created.

5.1 Dashboard

The **Dashboard** is your landing page. It summarises recent activity and the processing pipeline, and includes a **My Courses** card focused on the courses you own.

5.2 Creating a course

1. Select **My Courses** → **Create Course**.
2. Fill in:
    - **Title**
    - **Description**
    - **Category**
    - **Cover image** (optional) — a PNG, JPEG or WebP image up to 2 MB. If you do not provide one, ClassQuest draws a category-tinted placeholder.
3. Select **Save as draft**. The course is created in **Draft** status, so students cannot see it yet.

5.3 Adding a resource

Open one of your courses, then:

1. Select **Add Resource**.
2. Choose the type: **Document**, **Book** or **Video**.
3. Enter a **title**, an optional **description**, and an optional **week or section label** (used to group resources on the course page).
4. Drag and drop a file, or browse for one. Keep demonstration files small.
5. Select **Upload & publish**.

A **Processing pipeline** indicator moves through **Submitted** → **Queued** → **Processing** → **Completed**, usually within a few seconds. When it reaches **Completed**, the resource joins the course's

ordered list.

> Only the file **type and size** are checked. If a file is rejected, nothing is stored and you see a short message explaining why — just try a different file.

5.4 Arranging resources

On the course page you can:

- **Reorder** resources with the up/down arrows. New resources are added to the end of the list.
- **Edit** a resource's details inline.

5.5 Publishing, archiving and restoring a course

A course moves through a simple lifecycle. Use the controls on the course page (or **Edit Course**):

- **Publish Course** — moves a **draft** to **published**. Students can now see the course and its completed resources.
- **Return to draft** — moves a **published** course back to **draft**, hiding it from students again.
- **Archive course** — hides the course from students immediately. Its resources and access history are kept, but you cannot add new resources while it is archived.
- **Restore to draft** — brings an **archived** course back to **draft** so you can work on it again. (An archived course cannot go straight back to published; restore it to draft first, then publish.)

Allowed transitions at a glance:

From	You can go to
Draft	Published, Archived
Published	Draft, Archived
Archived	Draft

5.6 A note on what courses do and do not include

Courses are deliberately lightweight. There is **no enrolment** — every student sees every published course. There are no grades, quizzes, assignments, forums or certificates. Resource ordering uses the up/down arrows rather than drag-and-drop.

Chapter 6

Administrator guide

Administrators focus on the health of the system and on browsing the full library. Administrators can also manage any course (not just their own) through the system's interface.

6.1 Operations

The **Operations** screen is the control centre for a demonstration. It shows the live state of the system and provides the demonstration controls described in Demonstration controls.

Key panels:

- **Service health** — the status of each underlying service the system depends on (database, object storage, message queue, monitoring and alerting). If something is down, it shows here.
- **Storage overview** — how resource files are distributed across storage tiers.
- **Monitoring** — the count of recent bad (HTTP 400) requests and the monitoring alarm state.

6.2 Library

The **Library** gives administrators a view of **every** resource across all courses, which is useful for an overview or spot checks. (Teachers also have a library view limited to the all-resources list; students instead use **Courses**.)

Chapter 7

Demonstration controls

The **Operations** screen (available to teachers and administrators) includes a set of **Demonstration Controls**. These exist to show how the system behaves; everything they create is clearly labelled as demo/sample content so students never see it by accident.

Control	What it does
Seed demo catalogue	Creates four sample courses — *Cloud Computing*, *Applied Blockchain*, *Data Analytics* (published) and *Cybersecurity Essentials* (draft) — and uploads eleven resources through the normal pipeline. Running it again skips anything that already exists.
Induce processing failure	Deliberately makes one resource fail processing to show automatic retries. After the third attempt the resource is marked **Failed** and the message is moved to a dead-letter queue. The failing resource stays in a hidden draft course, so students never see it.
Simulate Glacier tiering	Moves up to ten completed demo resources to the long-term "Glacier" storage tier immediately, instead of waiting for the normal 90-day rule.
Generate HTTP 400 burst	Sends a burst of deliberately bad requests to demonstrate monitoring. The monitoring panel switches to **Threshold exceeded** and shows the count.

> **Local demonstration limits.** When running locally, a few cloud behaviours are simulated rather than fully reproduced: the monitoring **alarm** is not expected to change to the fired state, alert **emails** are never actually sent, and **Glacier restore** is not implemented (so opening a resource that has been moved to the Glacier tier may be refused — which is how real long-term storage behaves without a restore step). These are expected and are labelled in the interface.

Chapter 8

Running ClassQuest locally

This section is for the person setting up a demonstration. It summarises the quickest path to a running system. For full detail, Windows PowerShell commands and troubleshooting, see the companion document `docs/LOCAL_SETUP.md`; for a scripted walkthrough, see `DEMO.md`.

8.1 What you need

- **Docker Desktop** (with Docker Compose), running.
- **Node.js 20 or later** — only needed on the host to run the automated tests and helper scripts; the application itself runs inside Docker.
- The network ports **8080, 4000, 3306 and 4566** free.
- No AWS account and no cloud cost — the cloud services are emulated locally.

8.2 Start the stack

From the project folder:

```
# 1. Create your local settings file (defaults work as-is)
cp .env.example .env          # Windows PowerShell: Copy-Item .env.example .env

# 2. Build and start everything
docker compose up -d --build

# 3. Wait until the app-tier and web-tier show "healthy"
docker compose ps
```

The first build takes a few minutes. Then open **http://localhost:8080** and sign in with one of the demonstration accounts.

8.3 Seed the sample content

With the stack running, create the sample courses and resources either from the interface (**Operations → Demonstration Controls → Seed demo catalogue**, as a teacher or administrator)

or from the command line:

```
npm run demo:seed
```

8.4 Stop or reset

```
docker compose down        # stop, but keep all data
docker compose down -v     # stop and DELETE all data (fresh start next time)
```

> **Warning:** `docker compose down -v` permanently deletes the local database and stored files. Use plain `docker compose down` if you want to keep your demo data for next time.

Chapter 9

Troubleshooting

Symptom	Likely cause and fix
The page will not load at `localhost:8080`	The stack may still be starting. Run `docker compose ps` and wait until `app-tier` and `web-tier` are **healthy**.
"Docker daemon" or "engine" errors	Docker Desktop is not running. Start it and wait for *Engine running*, then start the stack again.
A container will not start because a port is "already allocated"	Another program is using port 8080, 4000, 3306 or 4566 (often a local MySQL on 3306). Stop that program or free the port. See `docs/LOCAL_SETUP.md` for exact commands.
A student cannot see a course	The course is in **Draft** or **Archived** status, or its resources have not finished processing. Publish the course and wait for resources to reach **Completed**.
A resource shows but will not open	The download link may have expired (they last about five minutes) — select **Open** again. If the resource was moved to the Glacier tier during a demo, it may be refused by design.
A newly uploaded resource is missing	Processing may still be running. Wait a few seconds for **Submitted** → **Queued** → **Processing** → **Completed**, or check the pipeline indicator.
The monitoring alarm never "fires" during a demo	Expected when running locally — the local emulator does not evaluate the alarm. The request count still updates and is labelled accordingly.
You were signed out unexpectedly	Sign-in tokens expire after about an hour. Simply sign in again.

Chapter 10

Frequently asked questions

Is my data real? No. ClassQuest is a prototype that uses only synthetic sample content. It is intended for demonstration, not for managing real courses or student records.

Does "My Progress" mean my grade? No. Progress records which resources you have *opened*. It is not a grade, score, or measure of understanding or completion.

Why can't students enrol in a specific course? By design there is no enrolment. Every student can see every published course.

Can a teacher edit another teacher's course? No. Teachers manage only the courses they created. Administrators can manage any course.

Where can I learn how the system is built? See the project's `ARCHITECTURE.md` and `README.md`. Those documents describe the technical design and how the prototype maps to a full cloud deployment.

ClassQuest is a research proof-of-concept. All content is synthetic DEMO/SAMPLE data.